

Verification of whether the final matrix is in echelon form

1. Row 3 is a zero row. Row 1 and Row 2 are non-zero rows. $\therefore$ The condition that all non-zero rows must be above the zero rows is True. $\square$
2. Row 1 Leading Entry = 1 in Column 1
Row 2 Leading Entry = 1 in Column 2
The condition that each row has its leading entry in a column to the right of the columns of the leading entries of all rows above it is True.
3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0

The condition that each pivot column has only zeros in all such positions below the pivot position is True. $\square$

$\therefore$ The matrix $\begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & 1 & 11 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is in echelon form $\square$

There are no rows in the matrix of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$. $\therefore$ By Theorem 2, the system of equations corresponding to the augmented matrix $\begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & 1 & 11 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is consistent, i.e. there exists a solution.

$\therefore$ The solution of $AX=b$ is present $\square$

Claim: For the given A and c , $AX=c$ has ^{no} solution.

Proof: By Theorem 3, a solution of $AX=\hat{c}$ is the same as the solution of the system of equations corresponding to the augmented matrix

$$\begin{bmatrix} 1 & 2 & -1 & 1 \\ 4 & 9 & 7 & 2 \\ 5 & 11 & 6 & 0 \end{bmatrix}$$

Therefore, we need to show that this matrix corresponds to a system of equation which has no solution.

$$\begin{bmatrix} 1 & 2 & -1 & 1 \\ 4 & 9 & 7 & 2 \\ 5 & 11 & 6 & 0 \end{bmatrix} \xrightarrow[\text{Row 2} - 4 \times \text{Row 1}]{\text{Row 2} =} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 11 & -2 \\ 5 & 11 & 6 & 0 \end{bmatrix} \xrightarrow[\text{Row 3} - 5 \times \text{Row 1}]{\text{Row 3} =} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 11 & -2 \\ 0 & 1 & 11 & -5 \end{bmatrix}$$